

TopNET



Reference Manual



TopNET Reference Manual

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Rev. B

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Preface

Thank you for purchasing your Topcon receiver, survey product or accessory (the “Product”). The materials available in this manual (the “Manual”) have been prepared by Topcon Positioning Systems, Inc. (“TPS”) for owners of Topcon products. This Manual is designed to assist owners with the use of software (the “Software”) to be used with the Product and its use is subject to these terms and conditions (the “Terms and Conditions”).



NOTICE

Please read these Terms and Conditions carefully.

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Manual Conventions

This manual uses the following conventions:

Example	Explanation
File ► Exit	Click the File menu and click Exit.
Enter	Click the button labeled Enter.
Topo	Indicates the name of a dialog box or screen.
<i>Notes</i>	Indicates a field on a dialog box or screen, or a tab within a dialog box or screen.



TIP

Supplementary information that can help you configure, maintain, set up, or use a system.

**NOTICE**

Supplementary information that can have an affect on system operation, system performance, measurements, personal safety.

**CAUTION**

Notification that an action has the potential to adversely affect system operation, system performance, data integrity, or personal health.

Notes:

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

TopNET CORS

Welcome to TopNET CORS, Topcon's unique and powerful software package designed to offer the easiest and most reliable method of managing any number of Continuously Operating Reference Stations (CORS) which takes advantage of the full range of satellite data available including both the GPS and GLONASS systems. TopNET CORS is provided as a series of modules, facilitating both ease of operation as well as future expansion to TopNET RTK and TopNET+ as your requirements for enhanced functionality increase.

TopNET CORS offers the following key features for the operation of from one to many CORS sites:

- Communication to Reference Stations via internet, GSM and cable modem, USB or serial port;
- Configuration of Reference Stations including remote firmware and Option Authorization File (OAF) upload;
- Configuration of Site parameters;
- Configuration of DGPS/RTK local broadcast from each Reference Station;
- Configuration of data logging for each Reference Station;
- Automatic and manual file download from each Reference Station;
- Automatic push to FTP site(s) of raw and/or RINEX data files;
- Monitoring of satellite and SNR (Signal-to-Noise Ratio);
- Displaying the current navigational coordinates Reference Stations, the number of tracked satellites (GPS and GLONASS) and current receiver time;
- Automated message sending on alarm via e-mail.

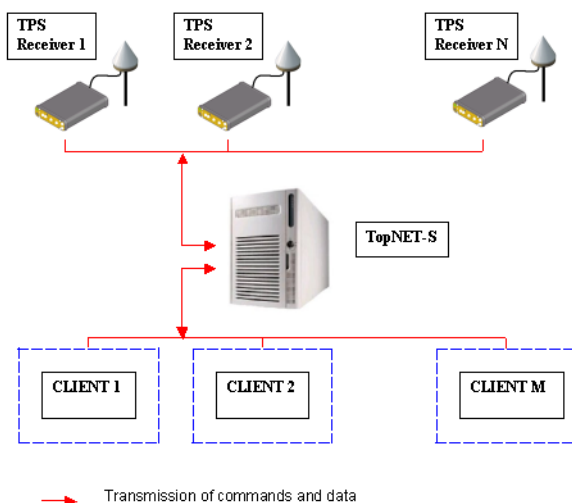


Figure 1-1. TopNET CORS Configuration

A company or agency offering GPS+ reference station services to their own employees or the surrounding community demand system flexibility without sacrificing security. With TopNET CORS you can log, download, and upload both raw data files and RINEX converted files to multiple FTP sites, all under the security and control of a designated System Manager. The System Manager has full rights to control remote receivers, and can give full rights or subset of the controls to users in the field. The System Manager has full control over all aspects of the receiver and data logging, including the ability to remotely load new firmware into CORS receivers, all from any computer with sufficient access rights.

With TopNET CORS you can also use the CORS stations to provide real-time correction data to support real-time DGPS and RTK users using standard RTCM and CMR data formats. Multiple communications options are available for use depending on local regulations, system operator requirements, user requirements and available data communication services. These various communications options are described in separate sections of this Manual as well as through continual updating of communications options provided by Topcon as new services become available. Please

contact your local Topcon distributor for the latest information on communications options available in your area.

TopNET CORS is designed to be operated as a client and server architecture using the two modules TopNET-S (server) and TopNET-R (client). TopNET-S manages the communication to the CORS receiver(s) in the network of CORS sites. Both TopNET-S and TopNET-R require the presence of the supplied Hardware Protection Keys to operate. One key will allow for operation of TopNET-S and TopNET-R on the same computer and the other key will allow the use of TopNET-R on an additional computer. Note that TopNET-S and TopNET-R can be installed on as many computers as desired and additional Hardware Protection Keys are available to enable the operation of the software on additional computers. Every effort has been made to insure that this will not interfere with continuous system operation and an unlimited number of CORS receivers and FTP servers are allowed. The use of the Hardware Protection Keys is described more fully in “Hardware Protection Key” on page 1-12.

Installation

Technical Specifications

Minimum Computer specifications are as follows:

Table 1-1. Computer Requirements for TopNET CORS

TopNET CORS Server with TopNET-S and TopNET-R	TopNET CORS Client with TopNET-R
• Processor compatible with Intel Pentium 1.5 GHz or faster • 1 GB free disk space • Minimum 512 MB RAM • MS Windows XP or Windows Server 2003 • Internet access with a fixed IP address • Color Monitor with minimum 640x480 screen resolution	• Processor compatible with Intel Pentium 1 GHz or faster • Minimum 1 GB free disk space, depending on data storage requirements • Minimum 128 MB RAM • MS Windows XP or Windows Server 2003 • Internet access • Color Monitor with minimum 640x480 screen resolution

Installing TopNET CORS on the Server Computer

Computer requirements are specified in “Technical Specifications” on page 1-4 of this Manual. Prior to installation of the TopNET CORS software, CORS sites should be fully installed and operational (See “Configuration of Continuously Operating Reference Station (CORS) sites” on page 1-13). At any time additional CORS stations may be incorporated into the network of CORS sites by following the instructions in the TopNET User’s Manual. Note that this requires the use of the Topcon software program PC-CDU which is supplied on the TopNET CD to configure the Topcon receiver for proper CORS operation.

TopNET CORS software comes on a CD to install on the computers. Both TopNET-S Server and TopNET-R client are on the CD as well as the manuals and the Topcon program PC-CDU which is used to configure Topcon GPS+ receivers for use in CORS networks.

1. Uninstall any old version of TopNET CORS manually. This can be done using **Add or Remove Programs** tool in **Control Panel**.
2. Insert the TopNET CD into the CD-ROM drive. The InstallShield Wizard starts up.
3. Click **Next** on the TopNET CORS install Wizard dialog box to start the installation process (Figure 1-2)

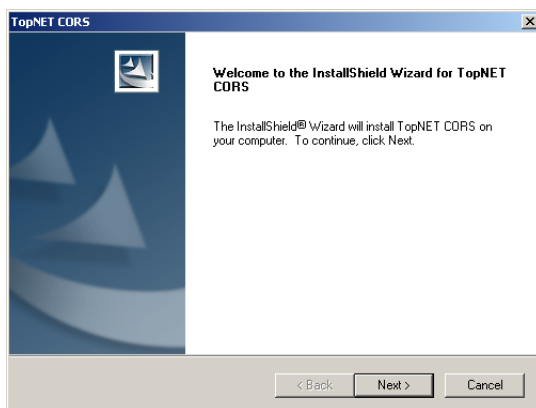


Figure 1-2. InstallShield Wizard dialog box for TopNET CORS

4. Click **Yes** on the License Agreement dialog box (Figure 1-3). Clicking **No** terminates the installation.

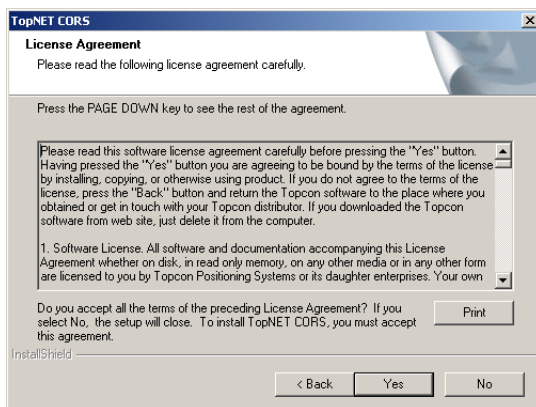


Figure 1-3. License Agreement

5. On the Choose Destination Location dialog box, click **Next** to select the default folder of *C:\Program Files\Topcon\TopNET* or click **Browse** to select a different folder name if desired. Click **Next** when desired folder is displayed (Figure 1-4).

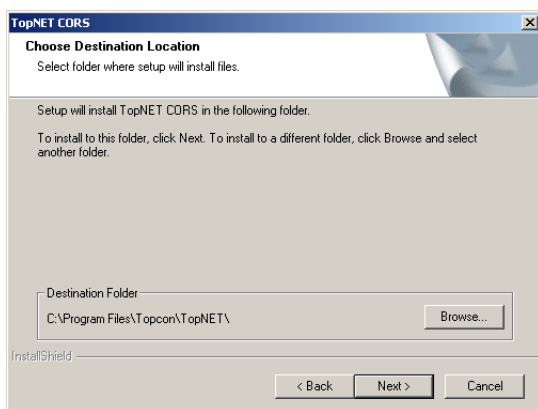


Figure 1-4. Choose Destination Location

6. On the Select Features dialog box, check those items you wish to install. Check all items for the TopNET CORS server computer and all except TopNET-S for client computers. Then click Next (Figure 1-5).

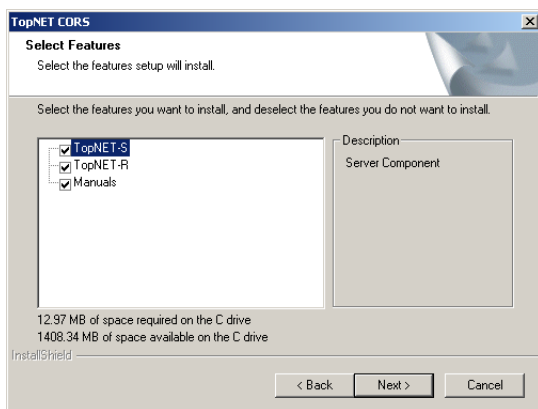


Figure 1-5. Select the features to install

- On the Select Program Folder dialog box, select a current folder or type a new program folder name for TopNET CORS (Figure 1-6).

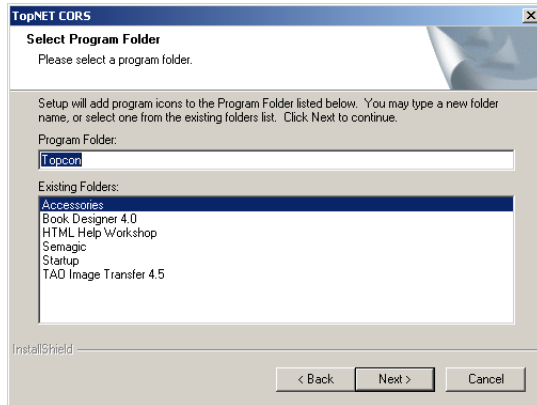


Figure 1-6. Select Program Folder

- The installation process (Figure 1-7) starts.

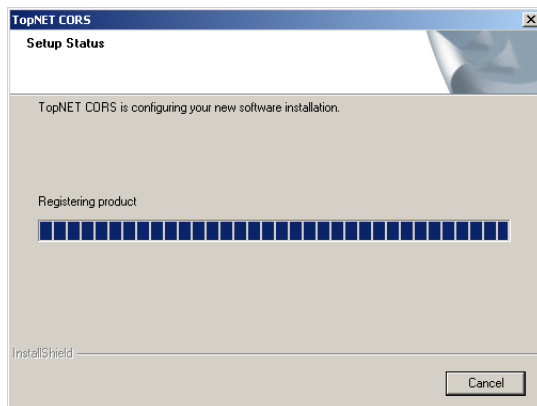


Figure 1-7. Installation in Progress

9. Click Finish to exit the installation (Figure 1-8).

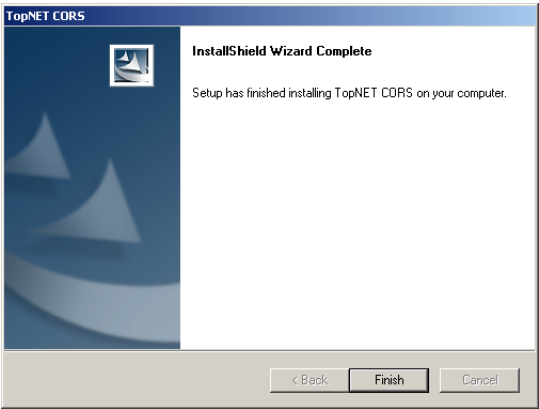


Figure 1-8. Installation Complete

The installation procedure places icons for all installed components (Table 1-2), as well as icons for associated manuals, on the computers desktop screen:

Table 1-2. TopNET CORS Program Icons

Program	TopNET-S	TopNET-R
Icon		

Installing TopNET-R on a Client Computer

TopNET-R can be installed on any number of computers, without installing TopNET-S.

1. Insert the TopNET CD into the CD-ROM drive of the client system. The InstallShield Wizard starts up.
2. Follow the on-screen instructions for installing TopNET CORS programs.

- On the **Select Features** dialog box, deselect the TopNET-S component. Make sure the TopNET-R component and Manuals are selected, then click **Next** (Figure 1-9).

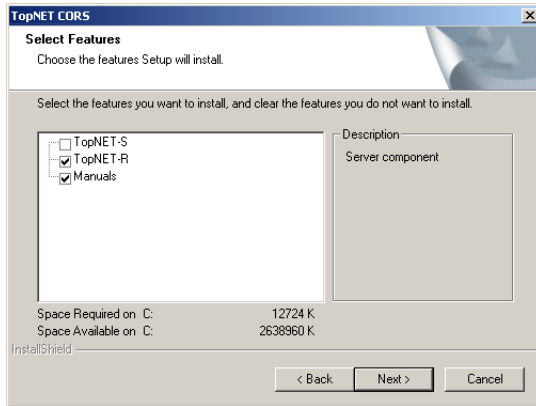


Figure 1-9. Select Features to Install – Client

- Continue with the installation.

When finished, TopNET CORS places a TopNET-R and Manual icons on the computer's desktop.



NOTICE

TopNET-R requires a Hardware Protection Key to operate on client systems. The TopNET CORS package includes one key for client systems to operate only TopNET-R.

Starting TopNET CORS

Following the installation of TopNET CORS, either the TopNET-S or the TopNET-R or both, icons will be on the computer desktop screen. If PC-CDU has been installed, an icon for this program will also appear.

Prior to operation, all Reference Stations should be installed at suitable CORS sites and the receiver(s) configured properly with Topcon PC-CDU (see TopNET User's Manual). Once all sites are ready for operation, TopNET-S is used to add CORS sites to the

Network as defined by the system administrator. Additional Reference Stations (CORS sites) may be added and incorporated into one or many Networks, or removed any time as defined by the system administrator.

The TopNET CORS server system requires a provided Hardware Protection Key. Insert this key into any USB port of the computer.

The TopNET CORS package consists of two modules:

- TopNET-S is used to establish the connection with the Reference Stations (CORS sites) and must be started to make the TopNET CORS operate. TopNET-S also registers TopNET-R clients and assigns rights for each client. Detailed instructions for the TopNET-S operation can be found in the TopNET User's Manual.
- TopNET-R is used to control the Reference Stations in the network of CORS stations and can be run as a client from any number of computers that have TopNET-R installed and a Hardware Protection Key present. TopNET-R also controls the download of data logged at each CORS site, the conversion of raw data to the RINEX format, and the uploading of both raw and RINEX data to the FTP site(s). Detailed instructions for the TopNET-R operation can be found in the TopNET User's Manual.

To start a TopNET CORS server do either of the following:

- Click **Start->Programs->Topcon->TopNET-S**, or
- Double-click the TopNET-S shortcut icon  on the computer desktop.

To start TopNET CORS client, TopNET-R, do either of the following

- Click **Start->Programs->Topcon->TopNET-R**, or
- Double-click the TopNET-R shortcut icon  on the computer desktop display.

TopNET Launcher

The TopNET Launcher service is installed along with TopNET package as an Automatic Startup service. It launches TopNET-S at PC startup if this application was running during the latest computer shutdown. Also the Download Center application of the TopNET-R client will be launched if it was not stopped or closed before the shutdown.

The status of the TopNET Launcher service can be seen in the list of available services of the computer.



NOTICE

Applications launched by TopNET Launcher may not appear in a Remote Desktop session. To make TopNET windows appear in a Remote Desktop session, launch the Remote Desktop client as a console terminal. For that purpose type in a command line:

`mstsc /console <full path to .rdp file>`

Only Windows Server 2000/2003/2008 systems support multiple user sessions. Remote Desktop service to an XP computer will connect you to the only session on that machine, and you will see all applications running there (including TopNET), so “console” switch is not needed if your server machine is running under Windows XP. If your client computer (the one FROM which you are launching Remote Desktop) is running Windows Vista Service Pack 1 or Windows XP Service Pack 3, “/console” switch will not work. This switch is not supported in the new version of mstcs coming with these Service Packs¹. For details refer to corresponding section² of the Microsoft Windows XP Help and Support Center.

-
1. However you can try to use an undocumented switch “/admin”.
 2. To find it, enter “mstsc” in the Search field of Help and Support Center.

Hardware Protection Key

The Hardware Protection Key is required for all computers running the TopNET-S Server and TopNET-R Client programs. Typically, there is only one computer per system configured to construct multiple networks from a list of available Reference Stations. This computer requires the installation of both TopNET-S Server and TopNET-R Client and presence of a suitable Hardware Protection Key enabled for both programs.



NOTICE

One Hardware Protection Key which enables the use of TopNET-S and TopNet-R on the same computer is supplied as part of the TopNET CORS package.

The Hardware Protection Keys are designed to operate in conjunction with the TopNET CORS software to insure continuous operation of the software. The software detects the Hardware Protection Key every hour and resets the program to run without a key for 14 days following the latest check. In this manner, if the Hardware Protection Key fails for any reason, the system user will be notified and will have sufficient time to contact Topcon Positioning Systems representatives for a replacement key. The system can be configured to notify Topcon automatically via e-mail as well.

TopNET-R also requires the use of a Hardware Protection Key and one TopNET-R only Key is provided as part of the TopNET CORS package. Note that TopNET-R can be installed on any number of computers and additional Keys are available to enable the operation of the TopNET-R Client on more than one computer.

If any of TopNET applications do not find Hardware Protection Key, a message of the following type appears (Figure 1-10):

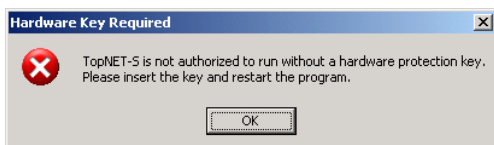


Figure 1-10. Warning: Hardware Key Required

Configuration of Continuously Operating Reference Station (CORS) sites

Continuously Operating Reference Stations provide a wide range of benefits to users of GPS, GLONASS and future GNSS information. To insure that this information, in the form of raw observations and information from the satellites about the satellites and systems, is of acceptable quality to support a wide range of users, certain guidelines must be followed to insure the proper operation of the CORS site(s).

As a minimum, CORS receivers need to be installed in a secure facility with provisions for a stable permanent mount for the antenna with regards towards providing a clear view of the sky with a minimum of obstructions that could block reception of satellite signals or nearby structures which could induce multipath errors in the satellite data. Furthermore, a CORS site should provide a continuous reliable source of power for the receiver as well as adequate means of communications between the CORS receiver and the site running the TopNET CORS software. This is best done via the internet using the Ethernet port available on Topcon Reference Stations although alternate means of communications include frame-relay, leased line or dial-up connection using the Ethernet, serial or USB ports with appropriate adapters and communications links.

Further information on CORS site requirements as well as information on the U.S. National CORS system and how to become a member of the National CORS or the Cooperative CORS networks can be found on the National Geodetic Survey site on the World Wide Web at <http://www.ngs.noaa.gov/CORS>.

Utilizing CORS sites to provide real-time DGPS and RTK data to mobile users

The individual CORS sites in TopNET CORS can be configured to provide real-time DGPS and / or RTK data to mobile users using a variety of communications options. As more communications options become available, information will be provided by Topcon Positioning Systems to enable users to take advantage of the latest technologies. Current available means of providing real-time data to mobile users include;

- **Radio** - A radio conforming to Topcon Positioning Systems requirements as well as local regulations may be attached to an open serial port on the Reference Station receiver. This radio will broadcast data to users equipped with compatible radios and provide DGPS and / or RTK operation over the area of radio coverage. Use of a radio provides a point to multi-point means of communications and will service an unlimited number of users within range of the broadcast radio.
- **CDMA** - A modem capable of operating within a CDMA network using IS-95B, a circuit switched connection, may be attached to an open serial port on the Reference Station receiver. This modem will provide DGPS and/or RTK operation to users equipped with compatible modems using wireless data communications. Use of a CDMA modem provides a point to multi-point means of communications and will service a number of users as long as CDMA coverage is available.
- **Mobile phone** - A mobile phone capable of operating within a GSM network using circuit switched data, may be attached to an open serial port on the Reference Station receiver. This mobile phone will provide DGPS and/or RTK operation to users equipped with compatible mobile phones using wireless data communications. Use of a mobile phone provides a point to point means of communications and will service a user as long as GSM coverage is available.

TopNET RTK

Welcome to TopNET RTK™, Topcon's unique and powerful software package designed to offer the easiest and most reliable method of managing Real Time Kinematics (RTK) surveys. RTK surveys take advantage of the full range of satellite data available, including both the GPS and GLONASS systems.

TopNET RTK offers the following key features for operating one, or many, reference stations:

- communication with Reference Stations via internet, GSM and cable modems, USB or serial ports;
- selection of the closest reference station to a rover receiver;
- real-time correction data transfer from reference stations to rover receivers;
- management of a large number of reference stations and mobile receivers simultaneously.

Figure 2-1 illustrates the layout of hardware and software connections in a TopNET RTK scheme with TopNET-N as a central data processor.

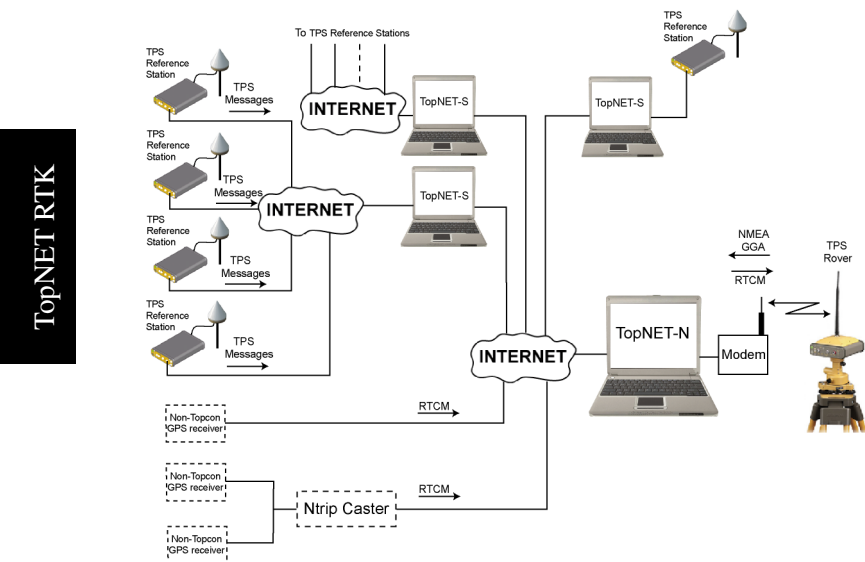


Figure 2-1. TopNET RTK Configuration

TPS Reference Stations are connected to a TopNET-S server via internet. Each TopNET-S server obtains data from one or more reference stations. The TopNET-N application collects RTCM data from TopNET-S servers and/or Ntrip casters via internet, processes them, and transmits to a mobile receiver using a data communication link.

A company or agency offering GNSS RTK services to their own employees or the surrounding community, demands system flexibility without sacrificing security. With TopNET RTK you can use the Reference Stations to provide real-time correction data to support real-time DGPS and RTK users using standard RTCM and CMR data formats. Multiple communications options are available for use depending on local regulations, system operator requirements, user requirements and available data communication services. Contact a local Topcon distributor for the latest information on communication options available in your area.

TopNET RTK is designed to operate as a client and server architecture using the TopNET-S (server) and TopNET-N (client) modules, as well as the TopNET-R client module for receiver and data control. TopNET-S manages the communication to the Reference Station receiver(s) in the network. TopNET-S, TopNET-R, and TopNET-N require the presence of the supplied Hardware Protection Keys to operate. The TopNET RTK package comes with two keys:

- one key will operate TopNET-S, TopNET-R, and TopNET-N on the same computer, and
- one key will operate TopNET-R on an additional computer.

The TopNET RTK modules can be installed to as many computers as desired. To enable the operation of the software on additional computers use the additional Hardware Protection Keys. Make sure that this will not interfere with continuous system operation, and an unlimited number of reference stations are permitted. See “Hardware Protection Key” on page 2-12 for more details on the keys.

Installation

TopNET RTK software comes on a CD to install on desired computers. TopNET-S, TopNET-R, and TopNET-N are on the CD, as well as all TopNET manuals. Topcon’s receiver configuration program, PC-CDU, is also available on the CD. PC-CDU configures Topcon GPS+ receivers for use in RTK networks.

Technical Specifications

The installation process of TopNET RTK package installs TopNET-S, TopNET-R, and TopNET-N applications.

These programs require the following minimum computer specifications to properly operate (Table 2-1).

Table 2-1. Computer Requirements for TopNET RTK

TopNET RTK Server with TopNET-S, TopNET-R, and TopNET-N	TopNET RTK Client with TopNET-R
• Processor compatible with Intel Pentium 1.5 GHz or faster • 1 GB free disk space • Minimum 512 MB RAM • MS Windows XP or Windows Server 2003 • Internet access with a fixed IP address • Color Monitor with minimum 640x480 screen resolution	• Processor compatible with Intel Pentium 1 GHz or faster • Minimum 1 GB free disk space, depending on data storage requirements • Minimum 128 MB RAM • MS Windows XP or Windows Server 2003 • Internet access • Color Monitor with minimum 640x480 screen resolution

Installing TopNET RTK on the Server Computer

See Table 2-1 for computer requirements for installing TopNET RTK (TopNET-S, TopNET-R, and TopNET-N) on the designated server computer.

Prior to installing TopNET RTK, reference stations should be fully installed and operational (see “Configuring Reference Stations for operation in RTK mode” on page 2-13). At any time, additional

reference stations may be incorporated into the network of Reference Stations using the procedures found in the TopNET User's Manual. Use Topcon's PC-CDU to configure the Topcon receiver for proper Reference Station operation.

1. Insert the TopNET CD into the CD-ROM drive or the server. The InstallShield Wizard starts up.
2. Click **Next** to start the installation process.

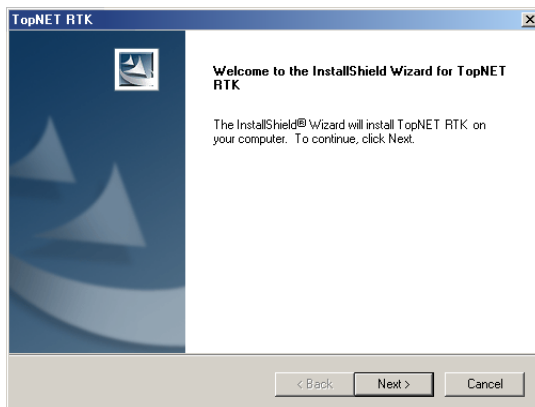


Figure 2-2. InstallShield Wizard Dialog Box for TopNET RTK

3. Click **Yes** on the **License Agreement** dialog box. Clicking **No** terminates the installation.

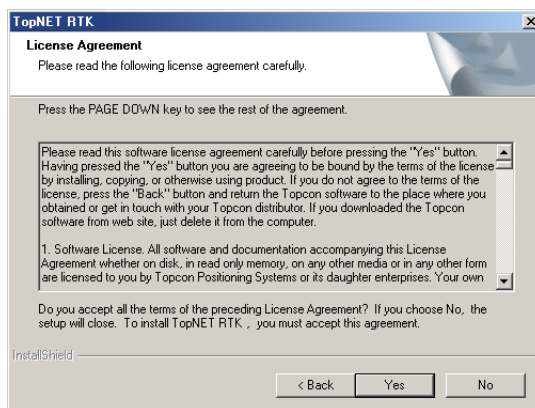


Figure 2-3. License Agreement

- On the **Choose Destination Location** dialog box, click **Next** to select the default folder (C:\Program Files\Topcon\TopNET) or click **Browse** to select a different folder. Click **Next** when the desired folder is displayed.

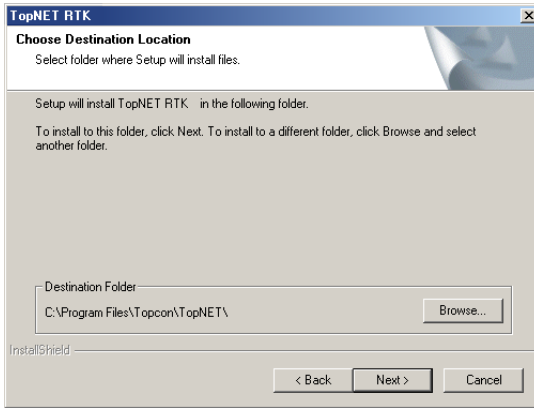


Figure 2-4. Choose Destination Location

- On the **Select Features** dialog box, select the features to install. Then click **Next**.

For the TopNET RTK server, select all items; for the TopNET-R client, select only the TopNET-R and Manuals items.

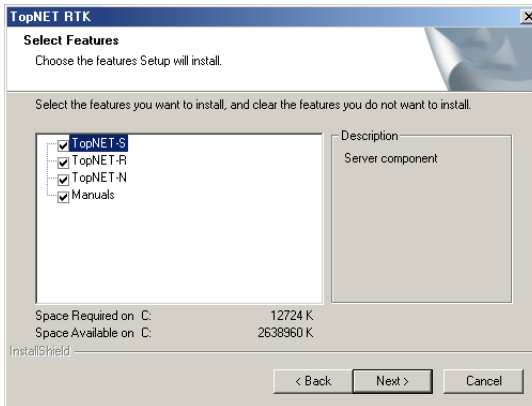


Figure 2-5. Select Features to Install

- On the **Select Program Folder** dialog box, select a current folder or type a new program folder name for TopNET RTK. Click **Next** to begin the installation.

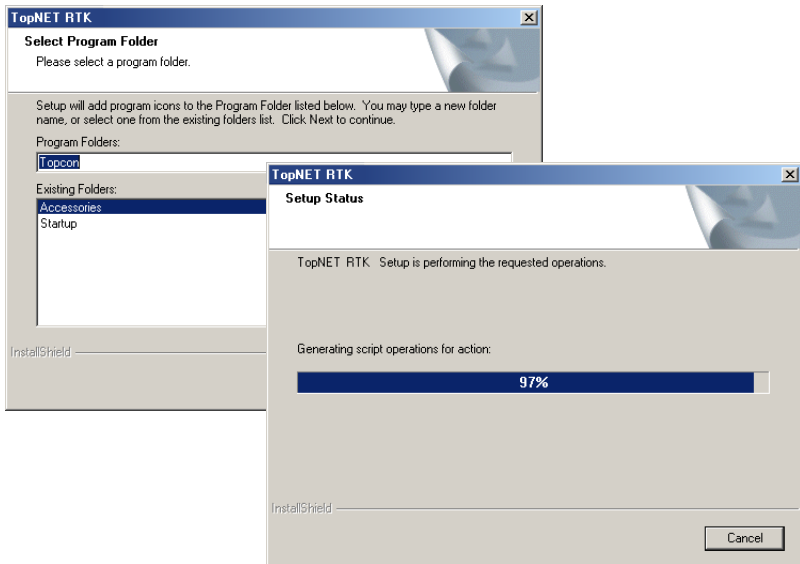


Figure 2-6. Select Program Folder and Installation Progress

- Click **Finish** to exit the installation.

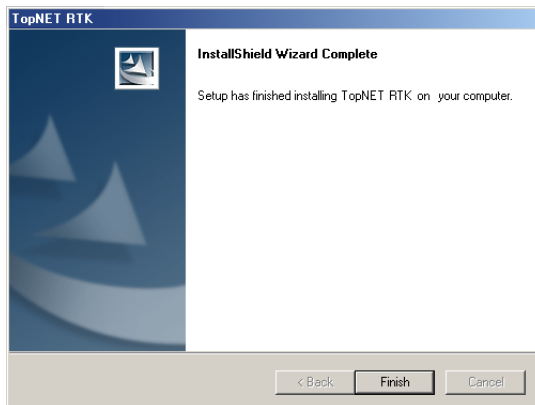


Figure 2-7. Installation Complete

The installation procedure places icons for all installed components (Table 2-2), as well as icons for associated manuals, on the computer’s desktop.

Table 2-2. TopNET RTK Program Icons

Program	TopNET-S	TopNET-R	TopNET-N
Icon			

NOTICE NOTICE

TopNET-S, TopNET-R, and TopNET-N require a Hardware Protection Key to operate on the server system. The TopNET RTK package includes one key for server systems that allows operation of all modules.

Installing TopNET-R on a Client Computer

TopNET-R can be installed on any number of computers, without installing TopNET-S.

1. Insert the TopNET CD into the CD-ROM drive of the client system. The InstallShield Wizard starts up.
2. Follow the on-screen instructions for installing TopNET RTK programs (step 2 on page 2-5 through step 5 on page 2-6).
3. On the **Select Features** dialog box, deselect the TopNET-S and TopNET-N components. Make sure the TopNET-R component and manuals are selected, then click **Next** (Figure 2-8).

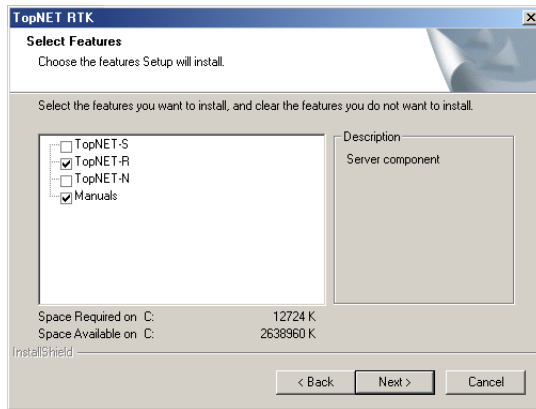


Figure 2-8. Select Features to Install – Client

4. Continue with the installation as described on the screen (steps 7 on page 2-7 through step 8 page 2-7).

When finished, TopNET RTK places TopNET-R and Manuals icons on the computer's desktop.



NOTICE

TopNET-R requires a Hardware Protection Key to operate on client systems. The TopNET RTK package includes one key for client systems that allows operation of TopNET-R only.

Starting TopNET RTK

Prior to operation, all Reference Stations should be installed at suitable Reference Station sites and the receivers configured properly with Topcon PC-CDU (refer to the TopNET User's Manual for details). Once all sites are ready for operation, use TopNET-S to add reference stations to the Network as defined by the system administrator. Additional Reference Stations may be added and incorporated into one or many Networks, or removed, any time as defined by the system administrator.

The TopNET RTK server and client systems require a Hardware Protection Key for operation. Before running any of the TopNET RTK applications, insert this key into an USB port of the computer.

The TopNET RTK package consists of three modules:

- TopNET-S establishes the connection with the Reference Stations and must be started to make the TopNET CORS operate. TopNET-S also registers TopNET-N clients and assigns rights for each client. Refer to the TopNET User's Manual for operating instructions.
- TopNET-N transfers real-time corrections data from the reference stations in the network of RTK stations. The application selects the best reference station for the rover receiver to receive these corrections from. Refer to the TopNET User's Manual for operating instructions.
- TopNET-R controls the Reference Stations in the network of CORS stations and can be run as a client from any number of computers that have TopNET-R installed and a Hardware Protection Key present. TopNET-R also controls the download of data logged at each CORS site, the conversion of raw data to the RINEX format, and the uploading of both raw and RINEX data to the FTP site(s). Refer to the TopNET User's Manual for operating instructions.

To start TopNET-S, the server component of TopNET RTK, click **Start ▶ Programs ▶ Topcon ▶ TopNET-S** or double-click the TopNET-S shortcut.

To start TopNET-N, the client component of TopNET RTK, click **Start ▶ Programs ▶ Topcon ▶ TopNET-N** or double-click the TopNET-N shortcut.

To start TopNET-R, the client component of TopNET CORS and TopNET RTK, click **Start ▶ Programs ▶ Topcon ▶ TopNET-R** or double-click the TopNET-R shortcut.

TopNET Launcher

The TopNET Launcher service is installed along with TopNET package as an Automatic Startup service. It launches TopNET-S and TopNET-N at startup if these applications were running during the latest computer shutdown. Also the Download Center application of the TopNET-R client will be launched if it was not stopped or closed before the shutdown.

The status of the TopNET Launcher service can be seen in the list of available services of the computer.



NOTICE

Applications launched by TopNET Launcher may not appear in a Remote Desktop session.

To make TopNET-S and TopNET-N windows appear in a Remote Desktop session, launch the Remote Desktop client as a console terminal. For that purpose type in a command line:

`mstsc.exe /console <full path to .rdp file>`

For details refer to corresponding section¹ of the Microsoft Windows XP Help and Support Center.

Following the instructions in the TopNET User's Manual, an initial network of from one to several Topcon Reference Stations can be added to multiple Network or "boot" configurations.

1. To find it, enter "mstsc" in the Search field of Help and Support Center.

Hardware Protection Key

All computers running TopNET-S server, TopNET-R client, and TopNET-N client require a Hardware Protection Key. An error message displays if the Hardware Protection Key is missing. Click **OK**, insert the key, and re-open the desired software module.

Typically, only one computer per system can be configured to construct multiple networks from a list of available Reference Stations, and this computer requires the installation of all three programs. A Hardware Protection Key enabled for all three programs must be inserted into the USB port to run the TopNET RTK modules on the server. Also the Hardware Protection Key limits the maximum number of reference stations that can be used in the TopNET-N application.

NOTICE

The TopNET RTK package includes one Hardware Protection Key that enables TopNET-S, TopNet-R, and TopNET-N on the same computer.

An additional Hardware Protection Key is included to run TopNET-R on an additional computer.

The Hardware Protection Keys are designed to operate in conjunction with the TopNET RTK software to ensure continuous operation of the software. The software detects the Hardware Protection Key every hour and resets the program to run without a key for 14 days following the latest check. In this manner, if the Hardware Protection Key fails for any reason, the system user will be notified and will have sufficient time to contact a Topcon Positioning Systems representative and swap the defective key.

NOTICE

Not more than four Hardware Protection Keys should be applied to one PC.

Configuring Reference Stations for operation in RTK mode

The Reference Stations provide a wide range of benefits to users of GPS, GLONASS, and future GNSS information. To ensure that this information, in the form of raw observations and information from the satellites about the satellites and systems, is of acceptable quality to support a wide range of users, certain guidelines must be followed to ensure the proper operation of the reference stations.

Reference station receivers need to be installed in a secure facility with provisions for a stable, permanent mount for the antenna that provides the following:

- A clear view of the sky.
- A minimum of obstructions that could block reception of satellite signals.
- A minimum of nearby structures which could induce multipath errors in the satellite data.
- A continuous, reliable source of power for the receiver.
- Adequate means of communications between the station and the site running the TopNET RTK software. Communication is best done via the internet using the Ethernet port available on Topcon Reference Stations. Alternate means of communications include frame-relay, leased line or dial-up connection using the Ethernet, serial or USB ports with appropriate adapters and communications links.

Visit the National Geodetic Survey web site for further information on reference stations requirements (<http://www.ngs.noaa.gov/CORS>), including information on the U.S. National CORS system and how to become a member of the National CORS or the Cooperative CORS networks.

Utilizing Reference Station Data for Mobile Users

The individual reference stations in TopNET RTK can be configured to provide real-time DGPS and/or RTK data to mobile users using a variety of communications options. As more communications options become available, information will be provided by Topcon Positioning Systems to enable users to take advantage of the latest technologies. Current options for providing real-time data to mobile users include the following:

- **Radio** – A radio conforming to Topcon Positioning Systems requirements and local regulations may be attached to an open serial port on the Reference Station receiver. This radio will broadcast data to users equipped with compatible radios and provide DGPS and/or RTK operation over the area of radio coverage. Use of a radio provides a point to multi-point means of communications and will service an unlimited number of users within range of the broadcast radio.
- **CDMA** – A modem capable of operating within a CDMA network using IS-95B, a circuit switched connection, may be attached to an open serial port on the Reference Station receiver. This modem will provide DGPS and/or RTK operation to users equipped with compatible modems using wireless data communications. Use of a CDMA modem provides a point to point means of communications and will service a limited number of users as long as CDMA coverage is available.
- **GSM mobile phone** – A mobile phone capable of operating within a GSM network using circuit switched data may be attached to an open serial port on the rover receiver. This mobile phone will provide DGPS and/or RTK operation to users equipped with compatible mobile phones using wireless data communications. Use of a mobile phone provides a point to point means of communications and will service a user as long as GSM coverage is available.

TopNET+

Welcome to TopNET+™, Topcon's unique and powerful software package designed to offer the easiest and most reliable method of managing surveys with Simulated Reference Station also called Network RTK. Network RTK surveys take advantage of the full range of satellite data available, including the GPS, GLONASS, and GALILEO systems and also the data from all the reference stations in the network.

TopNET+ offers the following key features for operating one, or many, reference stations;

- communication to Reference Stations via internet, GSM and cable modems, USB or serial ports
- selection of the closest reference station to the rover receiver
- real-time correction data transfer from reference stations to rover receivers
- management of a large number of reference stations and mobile receivers simultaneously
- construction of the corrections field that allows a rover to obtain the precise value of the corrections as if they were measured by a reference station located close to the Rover.

Figure 3-1 illustrates the connections between hardware and software with TopNET-V as the central data processor.

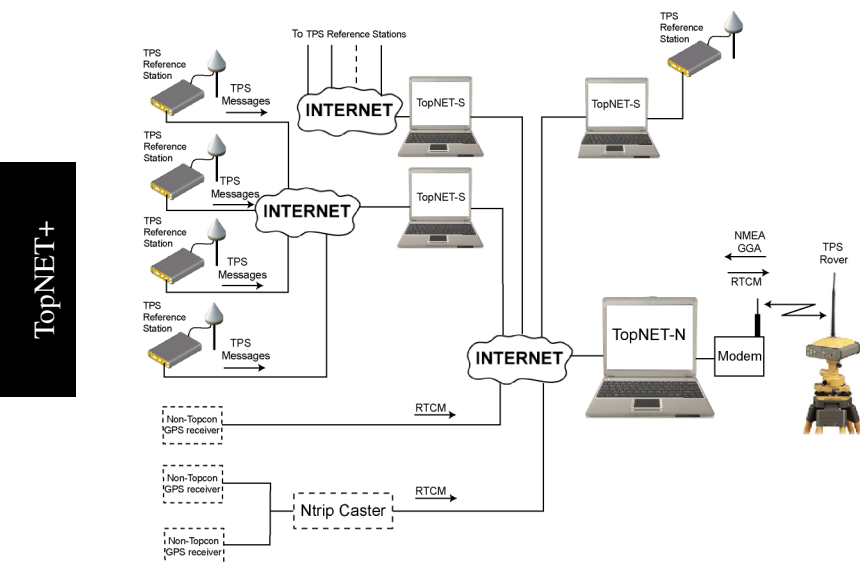


Figure 3-1. TopNET+ Configuration

TPS Reference Stations are connected to a TopNET-S server via the internet. Each TopNET-S server obtains data from one or more reference stations. The TopNET-V application collects RTCM data from TopNET-S servers and/or Ntrip casters via the internet, processes it, and transmits it to a mobile receiver using the data communication link.

A company or agency offering GNSS Network RTK services to their own employees or the surrounding community demands system flexibility without sacrificing security. With TopNET+ you can use the Reference Stations to provide real-time correction data to support real-time DGPS, RTK and Network RTK users using standard RTCM and CMR data formats. The only condition is that at least one of the reference stations should be Topcon manufactured. Multiple communications options are available for use depending on local regulations, system operator requirements, user requirements and available data communication services. Contact a local Topcon

distributor for the latest information on communications options available in your area.

TopNET+ is designed to operate as a client and server architecture using the TopNET-S (server) and TopNET-V (client) modules, as well as the TopNET-R client module for receiver and data control.

TopNET-S manages the communication to the Reference Station receiver(s) in the network. TopNET-S, TopNET-R, and TopNET-V require the presence of the supplied Hardware Protection Keys to operate. The TopNET+ package comes with two keys:

- one key will operate TopNET-S, TopNET-R, and TopNET-V on the same computer
- one key will operate TopNET-R on an additional computer

The TopNET+ modules can be installed on as many computers as desired, and additional Hardware Protection Keys are available to enable the operation of the software on additional computers. Every effort has been made to ensure that this will not interfere with continuous system operation and an unlimited number of reference stations are allowed. See “Hardware Protection Key” on page 3-12 for more details on the keys.

Installation

TopNET+ software comes on a CD to install on desired computers. TopNET-S, TopNET-R, and TopNET-V are on the CD, as well as all TopNET manuals. Topcon's receiver configuration program, PC-CDU, is also available on the CD. PC-CDU configures Topcon GNSS receivers for use in Network RTK networks.

Technical Specifications

The installation process for TopNET+ installs TopNET-S, TopNET-R, and TopNET-V. These programs require the following minimum computer specifications to properly operate (Table 3-1).

Table 3-1. Computer Requirements for TopNET+

TopNET+ Server with TopNET-S, TopNET-R, and TopNET-V	TopNET+ Client with TopNET-R
• Processor compatible with Intel Pentium 1.5 GHz or faster • 1 GB free disk space • Minimum 512 MB RAM • MS Windows XP or Windows Server 2003 • Internet access with a fixed IP address • Color Monitor with minimum 640x480 screen resolution	• Processor compatible with Intel Pentium 1 GHz or faster • Minimum 1 GB free disk space, depending on data storage requirements • Minimum 128 MB RAM • MS Windows XP or Windows Server 2003 • Internet access • Color Monitor with minimum 640x480 screen resolution

Installing TopNET+ on the Server Computer

See Table 3-1 for computer requirements for installing TopNET+ (TopNET-S, TopNET-R, and TopNET-V) on the designated server computer.

Prior to installing TopNET+, reference stations should be fully installed and operational (see "Configuring Reference Stations" on page 3-13). At any time, additional reference stations may be incorporated into the network of Reference Stations using the

procedures found in the TopNET User's Manual. Use Topcon's PC-CDU to configure the Topcon receiver for proper Reference Station operation.

1. Insert the TopNET CD into the CD-ROM drive of the server. The InstallShield Wizard starts up.
2. Click **Next** to start the installation process.

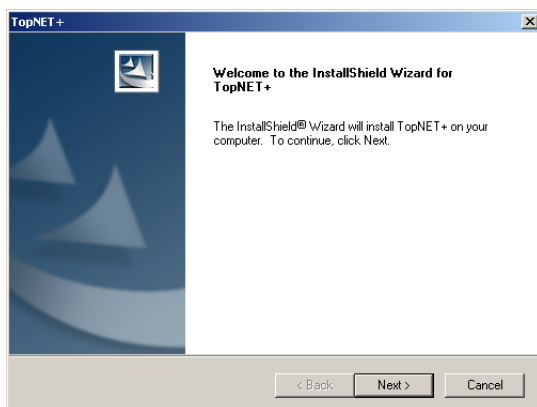


Figure 3-2. InstallShield Wizard Dialog Box for TopNET+

3. Click **Yes** on the **License Agreement** dialog box. Clicking **No** terminates the installation.

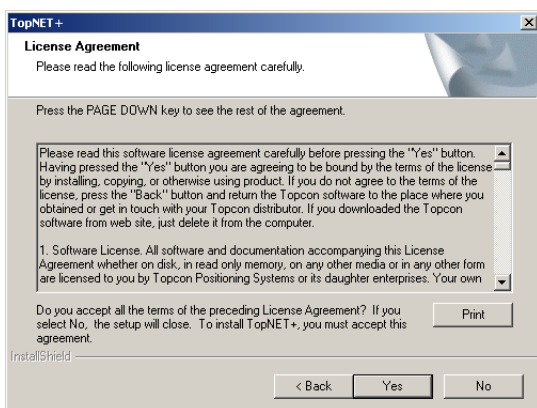


Figure 3-3. License Agreement

4. On the **Choose Destination Location** dialog box, click **Next** to select the default folder (C:\Program Files\Topcon\TopNET) or

click **Browse** to select a different folder. Click **Next** when the desired folder displays.

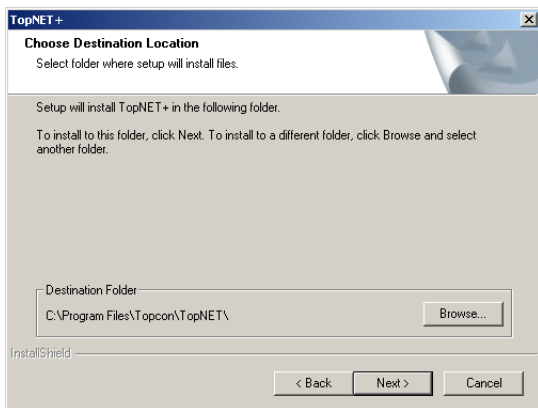


Figure 3-4. Choose Destination Location

5. On the **Select Features** dialog box, select the features to install. Then click **Next**.

For the TopNET+ server, select all items; for the TopNET-R client, select only the TopNET-R and Manuals items.

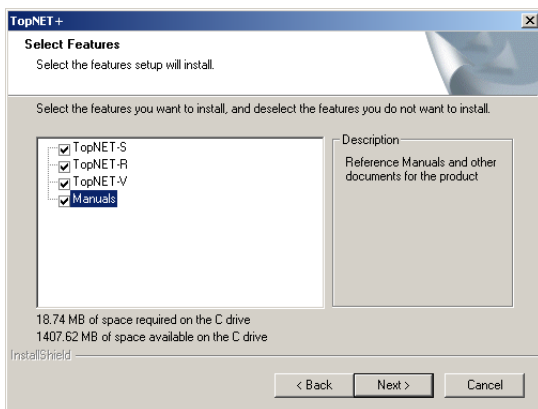


Figure 3-5. Select Features to Install

- On the **Select Program Folder** dialog box, select a current folder or type a new program folder name for TopNET+. Click **Next** to begin the installation.

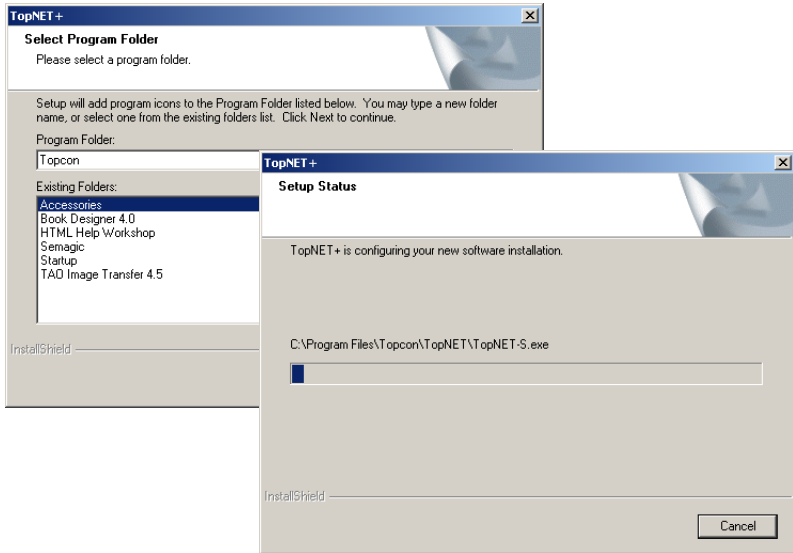


Figure 3-6. Select Program Folder and Installation Progress

- Click **Finish** to exit the installation.

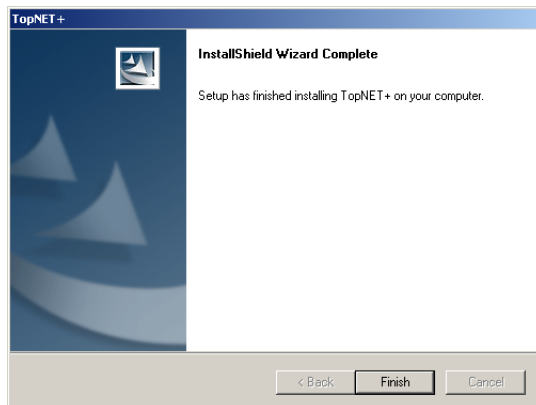


Figure 3-7. Installation Complete

The installation procedure places icons for all installed components (Table 3-2), as well as icons for associated manuals, on the computer’s desktop.

Table 3-2. TopNET+ Program Icons

Program	TopNET-S	TopNET-R	TopNET-V
Icon			

NOTICE NOTICE

TopNET-S, TopNET-R, and TopNET-V require a Hardware Protection Key to operate on the server system. The TopNET+ package includes one key for server systems to operate all modules.

Installing TopNET-R on a Client Computer

TopNET-R can be installed on any number of computers, without installing TopNET-S.

1. Insert the TopNET CD into the CD-ROM drive of the client system. The InstallShield Wizard starts up.
2. Follow the on-screen instructions for installing TopNET+ programs (step 2 on page 3-5 through step 5 on page 3-5).
3. On the **Select Features** dialog box, deselect the TopNET-S and TopNET-V components. Make sure the TopNET-R component and Manuals are selected and click **Next** (Figure 3-8).

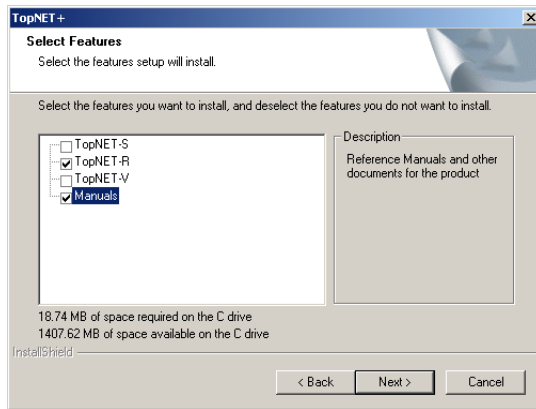


Figure 3-8. Select Features to Install – Client

4. Continue with the installation as described on the screen (steps 7 on page 3-7 through step 8 page 3-7).

When finished, TopNET+ places TopNET-R and Manual icons on the computer's desktop.



NOTICE

TopNET-R requires a Hardware Protection Key to operate on client systems. The TopNET+ package includes one key for client systems that allows operation of TopNET-R only.

Starting TopNET+

Prior to operation, all Reference Stations should be installed at suitable Reference Station sites and the receivers configured properly with Topcon PC-CDU (refer to the TopNET User's Manual for details). Once all sites are ready for operation, use TopNET-S to add reference stations to the Network as defined by the system administrator. Additional Reference Stations may be added and incorporated into one or many Networks, or removed, any time as defined by the system administrator.

The TopNET+ server and client systems require the use of the supplied Hardware Protection Key. Before starting any TopNET+ program, insert this key into an USB port of the computer.

The TopNET+ package consists of three modules:

- TopNET-S establishes the connection with the Reference Stations and must be running at all times for TopNET+ to operate. TopNET-S also registers TopNET-V clients and assigns rights for each client. Refer to the TopNET User's Manual for operating instructions.
- TopNET-R controls the Reference Stations in the network of CORS stations and can be run as a client from any number of computers. TopNET-R also controls the download of data logged at each CORS site, the conversion of raw data to the RINEX format and the uploading of both raw and RINEX data to the FTP site(s). Refer to the TopNET User's Manual for operating instructions.
- TopNET-V transfers real-time corrections data from the reference stations in the network of Network RTK stations. The application selects the best reference station (or stations) for the rover receiver to receive these corrections from. Refer to the TopNET User's Manual for operating instructions.

To start TopNET-S, the server component of TopNET+, click **Start ▶ Programs ▶ Topcon ▶ TopNET-S** or double-click the TopNET-S shortcut.

To start TopNET-R, the client component of TopNET CORS and TopNET+, click **Start ▶ Programs ▶ Topcon ▶ TopNET-R** or double-click the TopNET-R shortcut.

To start TopNET-V, the client component of TopNET+, click **Start ▶ Programs ▶ Topcon ▶ TopNET-V** or double-click the TopNET-V shortcut.

TopNET Launcher

The TopNET Launcher service is installed along with TopNET package as an Automatic Startup service. It launches TopNET-S and TopNET-V at startup if these applications were running during the latest computer shutdown. Also the Download Center application of the TopNET-R client will be launched if it was not stopped or closed before the shutdown.

The status of the TopNET Launcher service can be seen in the list of available services of the computer.



NOTICE

Applications launched by TopNET Launcher may not appear in a Remote Desktop session.

To make TopNET-S and TopNET-V windows appear in a Remote Desktop session, launch the Remote Desktop client as a console terminal. For that purpose type in a command line:

`mstsc.exe /console <full path to .rdp file>`

For details refer to corresponding section¹ of the Microsoft Windows Help and Support Center.

Following the instructions in the TopNET User's Manual, an initial network of from one to several Topcon Reference Stations can be added to multiple Network or "boot" configurations.

1. To find it, enter "mstsc" in the Search field of Help and Support Center.

Hardware Protection Key

All computers running TopNET-S server, TopNET-R client, and TopNET-V client require a Hardware Protection Key. An error message displays if the Hardware Protection Key is missing. Click **OK**, insert the key, and re-open the desired software module.

Typically, only one computer per system can be configured to construct multiple networks from a list of available Reference Stations, and this computer requires the installation of all three programs. A Hardware Protection Key enabled for all three programs must be inserted into the appropriate port to run the TopNET+ modules on the server. Also the Hardware Protection Key limits the maximum number of reference stations that can be used in the TopNET-V application.



NOTICE

The TopNET+ package includes one Hardware Protection Key that enables TopNET-S, TopNet-R, and TopNET-V on the same computer.

An additional Hardware Protection Key is included to run TopNET-R on an additional computer.

The Hardware Protection Keys are designed to operate in conjunction with the TopNET+ software to ensure continuous operation of the software. The software detects the Hardware Protection Key every hour and resets the program to run without a key for 14 days following the latest check. In this manner, if the Hardware Protection Key fails for any reason, the system user will be notified and will have sufficient time to contact a Topcon Positioning Systems representative for a replacement key.



NOTICE

Not more than four Hardware Protection Keys should be applied to one PC.

Configuring Reference Stations

The Reference Stations provide a wide range of benefits to users of GPS, GLONASS, and future GNSS information. To ensure that this information, in the form of raw observations and information from the satellites about the satellites and systems, is of acceptable quality to support a wide range of users, certain guidelines must be followed to ensure the proper operation of the reference stations.

Reference station receivers need to be installed in a secure facility with provisions for a stable, permanent mount for the antenna that provides the following:

- A clear view of the sky.
- A minimum of obstructions that could block reception of satellite signals.
- A minimum of nearby structures which could induce multipath errors in the satellite data.
- A continuous, reliable source of power for the receiver.
- Adequate means of communications between the station and the site running the TopNET+ software. Communication is best done via the internet using the Ethernet port available on Topcon Reference Stations. Alternate means of communications include frame-relay, leased line or dial-up connection using the Ethernet, serial or USB ports with appropriate adapters and communications links.

Visit the National Geodetic Survey web site for further information on reference stations requirements (<http://www.ngs.noaa.gov/CORS>), including information on the U.S. National CORS system and how to become a member of the National CORS or the Cooperative CORS networks.

Utilizing Reference Station Data for Mobile Users

The individual reference stations in TopNET+ can be configured to provide real-time DGPS and/or RTK data to mobile users using a variety of communications options. As more communications options become available, information will be provided by Topcon Positioning Systems to enable users to take advantage of the latest technologies. Current options for providing real-time data to mobile users include the following:

- **Radio** – A radio conforming to Topcon Positioning Systems requirements and local regulations may be attached to an open serial port on the Reference Station receiver. This radio will broadcast data to users equipped with compatible radios and provide DGPS and/or RTK operation over the area of radio coverage. Use of a radio provides a point to multi-point means of communications and will service an unlimited number of users within range of the broadcast radio.
- **CDMA** – A modem capable of operating within a CDMA network using IS-95B, a circuit switched connection, may be attached to an open serial port on the Reference Station receiver. This modem will provide DGPS and/or RTK operation to users equipped with compatible modems using wireless data communications. Use of a CDMA modem provides a point to point means of communications and will service a limited number of users as long as CDMA coverage is available.
- **GSM mobile phone** – A mobile phone capable of operating within a GSM network using circuit switched data may be attached to an open serial port on the rover receiver. This mobile phone will provide DGPS and/or RTK operation to users equipped with compatible mobile phones using wireless data communications. Use of a mobile phone provides a point to point means of communications and will service a user as long as GSM coverage is available.

[illegible]

Notes:

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



Topcon Positioning Systems, Inc.

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ISO 9001:2000
FM 68448

TopNET Reference Manual
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